Gokul Palanisamy

Software Engineer | Image Processing | Computer Vision | Machine Learning

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PROFESSIONAL SUMMARY

Professional with 3.2 years of software industry experience in Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, and Image Processing. Strong hands-on experience in Python, Java, and MySQL, with expertise in developing image processing algorithms, image segmentation, object detection, feature extraction, and classification models. Completed a Java Full-Stack Web Development professional training with a strong foundation in object-oriented programming, databases, and problem-solving. Passionate about applying artificial intelligence, software engineering, and research-driven approaches to solve real-world engineering challenges.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, MySQL
- **Image Processing:** OpenCV, NumPy, SciPy, Image Processing Toolbox
- **Machine Learning:** SVM, KNN, Random Forest, Feature Engineering
- **Deep Learning:** CNN, PyTorch TensorFlow, Deep Learning Toolbox
- **Tools & Cloud:** Git, VS Code, Docker, Kubernetes, Jenkins, JIRA, AWS (EC2, S3, IAM, RDS, Cloudwatch)

PROFESSIONAL EXPERIENCE

Client Acquisition Associate
Nov 2025 - Sep 2026

Elythra Edufyi Tech Solutions — *B*engaluru

- Led technical sales initiatives, guided the team in client engagement, solution selling, and contributed to 1.5 Cr in revenue generation.
- Designed and delivered workshops, webinars on image preprocessing, resizing, segmentation, and optimization techniques to train ML models on specific use cases.
- Collaborated with cross-functional teams to identify evolving industry requirements and align learning initiatives with relevant business needs and market trends.
- Researched emerging AI/ML frameworks to recommend suitable tech stacks for real-time projects of students.

Research Associate
Jul 2022 - Sep 2024

Imatiz Technologies — *B*engaluru

- Conducted research on AI-based medical image processing applications for disease diagnosis using machine learning and deep learning techniques.
- Implemented image segmentation, object detection, feature extraction, and classification algorithms.
- Examined and evaluated machine learning and deep learning models using Python.
- Worked closely with cross-functional teams to streamline workflows, improving project delivery by 20%.
- Mentored junior engineers and collaborated on multiple R&D projects, contributing to technical problem-solving and successful project execution.

PROFESSIONAL TRAINING

Java Fullstack Web Development
Nov 2024 - Oct 2025

Pankaj Sir Academy — *B*engaluru

- Designed and developed a fullstack application using Spring Boot following a layered MVC architecture with 10+ REST APIs for CRUD operations and user management.
- Implemented authentication and authorization using Spring Security and JWT-based mechanisms, including role-based access control for admin and user roles.

- Deployed the application on AWS EC2 and set up CI/CD pipelines using Jenkins and GitHub Actions to automate build, test, and deployment steps.
- Performed debugging, performance tuning, and unit testing to ensure application reliability and maintainability.

PROJECTS

Multi-Class Object Detection Using Drone Imagery - *Python, PyTorch, VisDrone, YOLO*

- Explored an object detection model using the VisDrone dataset to detect and classify vehicles from road images.
- Analyzed image preprocessing, data augmentation, and CNN-based feature extraction to prepare images for detection.
- Examined YOLO-based object detection with bounding-box localization and Non-Maximum Suppression (NMS) for evaluating detection performance.

Image Forgery Detection - *DWT, LBP*

- Conducted a research on image forgery detection algorithm using Discrete Wavelet Transform (DWT) and Local Binary Pattern (LBP) features.
- Analyzed texture inconsistencies and pixel-level artifacts for tampering detection.
- Delivered results on robustness against copy-move and image manipulation attacks.

Deep Learning-Based Facial Expression Recognition - *ImageNet, CNN, Computer Vision*

- Examined a CNN-based emotion classification model to detect human emotions from facial images.
- Analyzed face detection, image preprocessing, and data augmentation to improve feature quality and model robustness.
- Evaluated transfer learning with ImageNet-pretrained features and evaluated the model using accuracy, precision, recall, and F1-score.

HotelTonight - *Java, Spring Boot, Microservices, MySQL, Docker, JWT, Eureka, Feign Client*

- Developed a full-stack hotel booking application using Java, Spring Boot, Microservices, and MySQL.
- Built secure REST APIs with JWT authentication and explored Apache Kafka for event-driven communication, along with Eureka and Feign Client for service discovery and inter-service communication.
- Applied Docker, Spring Security, and Domain-Driven Design (DDD) concepts to build modular, secure, and scalable application services.

Employee Management System - *Java, Spring Boot, JPA, MySQL*

- Built a backend system using Java, Spring Boot, JPA, and MySQL with complete CRUD functionality.
- Implemented Role-Based Access Control (RBAC) to restrict operations based on user roles.
- Developed JWT-based authentication and authorization using Spring Security. Implemented secure password encryption and stateless session management.

EDUCATION

Velalar College of Engineering and Technology 2018 – 2022
Bachelor of Engineering in Electronics and Communication Engineering (ECE), CGPA - 8.42 Tamil Nadu

Kongu Vellalar Higher Secondary School 2017 – 2018
Higher Secondary Certificate (HSC), 78.6% Tamil Nadu

Kongu Vellalar Higher Secondary School 2015 – 2016
Secondary School Leaving Certificate (SSLC), 91.6% Tamil Nadu